Lab 3: Component Modelling & Architectural Pattern Selection

Self-Service Coffee Kiosk System - Architecture Justification

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System. The system is separated into a Presentation Layer for the touch-screen interface, a Business Layer for order, payment and receipt operations, and a Data Layer for menu and pricing storage.

Reason 1 - Simple fit for the kiosk requirements

The kiosk has a small and clearly defined workflow: choose a coffee and size, pay by credit card, and print a receipt. Layered Architecture keeps the user interface separate from the order-processing logic, so changes to the touch-screen screens do not require changes to payment, receipt, or database code.

Reason 2 - Clean separation of hardware and data access

The Order Manager acts as the central business component and communicates with the Payment Service, Receipt Printer, and Menu & Pricing Database through defined interfaces. This makes the hardware and data-related components easier to replace, test, and maintain without affecting the complete system.

Security Advantage

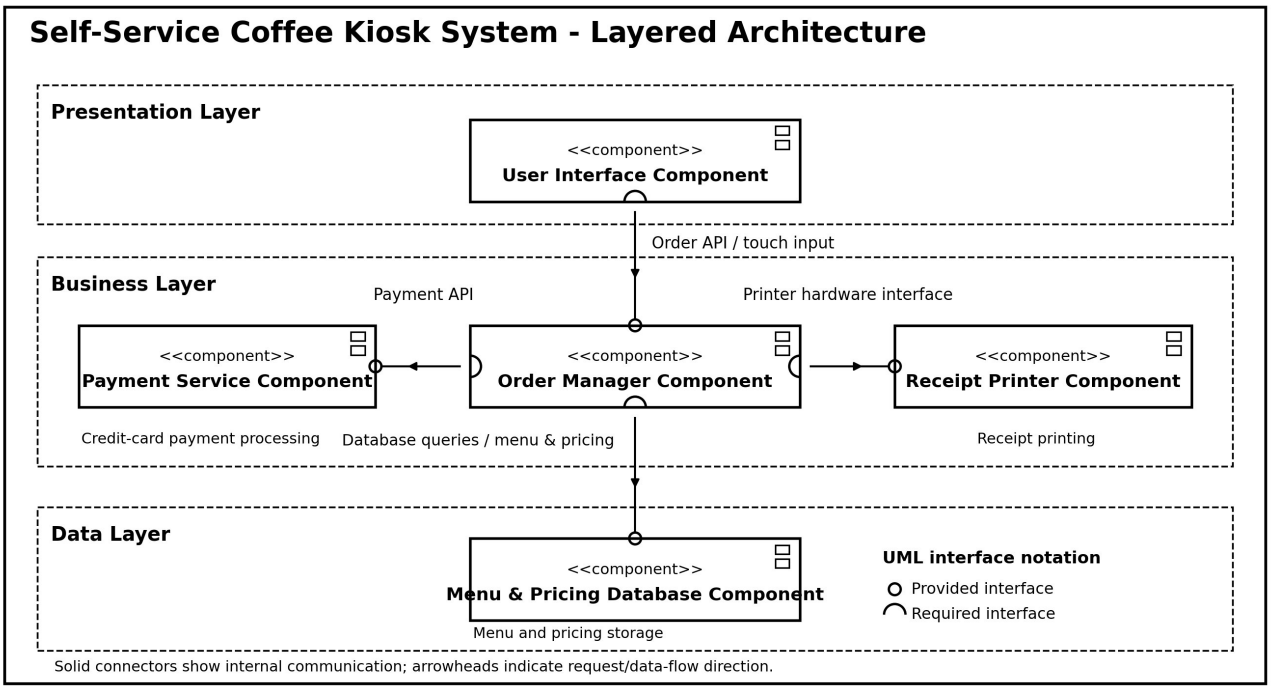
Payment functions are isolated inside the Payment Service instead of being handled directly by the user-interface component. This reduces the number of components that need access to card-processing functions and creates a clear boundary around payment-related operations.

Performance Benefit

For a single self-service kiosk, local communication between layers is lightweight and avoids the network and deployment overhead of a microservices design. Menu and pricing queries are also handled through one controlled path, keeping the ordering flow fast and predictable.

Component Mapping

Layer	Components
Presentation	User Interface Component
Business	Order Manager, Payment Service, Receipt Printer
Data	Menu & Pricing Database


Self-Service Coffee Kiosk System - Layered Architecture
Presentation Layer
<<component>>
User Interface Component
Order API / touch input
Business Layer
Payment API
Printer hardware interface
<<component>>
Payment Service Component
<<component>>
Order Manager Component
<<component>>
Receipt Printer Component
Credit-card payment processing
Database queries / menu & pricing
Receipt printing
Data Layer
<<component>>
Menu & Pricing Database Component
Menu and pricing storage
UML interface notation
Provided interface
Required interface
Solid connectors show internal communication; arrowheads indicate request/data-flow direction.